

UNITS

The course content is organized into commonly taught units. The units have been arranged in a sequence frequently found in many college courses and textbooks.

The eight units in AP Biology, and their weightings on the multiple-choice section of the AP Exam, are listed below.

Pacing recommendations at the unit level and on the Course at a Glance provide suggestions for how to teach the required course content and administer the Progress Checks. The suggested class periods are

based on a schedule in which the class meets five days a week for 45 minutes each day. While these recommendations have been made to aid in planning, adjust the pacing based on the needs of your students, alternate schedules (e.g., block scheduling), or your school's academic calendar.

TOPICS

Each unit is divided into teachable segments called topics. The topic pages (starting on p. 32) contain all required content for each topic.

Units	Exam Weighting
Unit 1: Chemistry of Life	8–11%
Unit 2: Cells	10–13%
Unit 3: Cellular Energetics	12–16%
Unit 4: Cell Communication and Cell Cycle	10–15%
Unit 5: Heredity	8–11%
Unit 6: Gene Expression and Regulation	12–16%
Unit 7: Natural Selection	13–20%
Unit 8: Ecology	10–15%

Spiraling the Big Ideas

The following table shows how the big ideas spiral throughout the course by showing the units in which each big idea appears.

Big Ideas		Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8
		Chemistry of Life	Cells	Cellular Energetics	Cell Communication and Cell Cycle	Heredity	Gene Expression and Regulation	Natural Selection	Ecology
Evolution			✓			✓		✓	✓
Energetics		✓	✓	✓	✓				✓
Information Storage and Transmission		✓			✓	✓	✓		✓
Systems Interactions		✓	✓			✓		✓	✓

THIS PAGE IS INTENTIONALLY LEFT BLANK.

Course at a Glance

Plan

The course at a glance provides a useful visual organization of the AP Biology components, including:

- Sequence of units, along with approximate weighting and suggested pacing. Please note, pacing is based on 45-minute class periods, meeting five days each week for a full academic year.
- Progression of topics within each unit.
- Spiraling of the big ideas and science practices across units.

Teach

SCIENCE PRACTICES

Science practices spiral throughout the course.

- | | |
|---------------------------------|--|
| 1 Concept Explanation | 4 Representing and Describing Data |
| 2 Visual Representations | 5 Statistical Tests and Data Analysis |
| 3 Questions and Methods | 6 Argumentation |

Assess

Assign the Progress Checks—either as homework or in class—for each unit. Each Progress Check contains formative multiple-choice and free-response questions. The feedback from the Progress Checks shows students the areas where they need to focus.

UNIT 1

Chemistry of Life

~9-11 Class Periods

8-11% AP Exam Weighting

- | | |
|----------|--|
| 2 | 1.1 Structure of Water and Hydrogen Bonding |
| 2 | 1.2 Elements of Life |
| 2 | 1.3 Introduction to Macromolecules |
| 1 | 1.4 Carbohydrates |
| 6 | 1.5 Lipids |
| 2 | 1.6 Nucleic Acids |
| 6 | 1.7 Proteins |

UNIT 2

Cells

~14-16 Class Periods

10-13% AP Exam Weighting

- | | |
|----------------------|--|
| 1
6 | 2.1 Cell Structure and Function |
| 2
5 | 2.2 Cell Size |
| 2 | 2.3 Plasma Membrane |
| 5 | 2.4 Membrane Permeability |
| 3 | 2.5 Membrane Transport |
| 6 | 2.6 Facilitated Diffusion |
| 4 | 2.7 Tonicity and Osmoregulation |
| 1 | 2.8 Mechanisms of Transport |
| 6 | 2.9 Cell Compartmentalization |
| 6 | 2.10 Origins of Cell Compartmentalization |

Progress Check 1

Multiple-Choice: ~24 questions

Free-Response: 2 questions

- Conceptual Analysis (partial)
- Analyze Model or Visual Representation of a Biological Concept or Process (partial)

Progress Check 2

Multiple-Choice: ~33 questions (2 parts)

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results (partial)
- Analyze Model or Visual Representation of a Biological Concept or Process (partial)

NOTE: Partial versions of the free-response questions are provided to prepare students for more complex, full questions that they will encounter on the AP Exam.

UNIT 3

Cellular Energetics

~12-14

Class Periods

12-16%

AP Exam Weighting

1
3

3.1 Enzymes

6

3.2 Environmental Impacts on Enzyme Function

6

3.3 Cellular Energy

6

3.4 Photosynthesis

4

3.5 Cellular Respiration

UNIT 4

Cell Communication and Cell Cycle

~12-14

Class Periods

10-15%

AP Exam Weighting

1

4.1 Cell Communication

1

4.2 Introduction to Signal Transduction

6

4.3 Signal Transduction Pathways

6

4.4 Feedback

4

5

4.5 Cell Cycle

6

4.6 Regulation of Cell Cycle

UNIT 5

Heredity

~8-10

Class Periods

8-11%

AP Exam Weighting

1

5.1 Meiosis

3

5.2 Meiosis and Genetic Diversity

5

6

5.3 Mendelian Genetics

5

5.4 Non-Mendelian Genetics

1

5.5 Environmental Effects on Phenotype

Progress Check 3

Multiple-Choice: ~19 questions

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results with Graphing (partial)
- Scientific Investigation (partial)

Progress Check 4

Multiple-Choice: ~24 questions

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results (partial)
- Analyze Data

Progress Check 5

Multiple-Choice: ~23 questions

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results with Graphing
- Conceptual Analysis

NOTE: Partial versions of the free-response questions are provided to prepare students for more complex, full questions that they will encounter on the AP Exam.

UNIT 6

Gene Expression and Regulation

~18-20 Class Periods **12-16%** AP Exam Weighting

- 1** 6.1 DNA and RNA Structure
- 2** 6.2 DNA Replication
- 2** 6.3 Transcription and RNA Processing
- 2** **6** 6.4 Translation
- 6** 6.5 Regulation of Gene Expression
- 6** 6.6 Gene Expression and Cell Specialization
- 2** 6.7 Mutations
- 6** 6.8 Biotechnology

Progress Check 6

Multiple-Choice: ~25 questions

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results
- Analyze Model or Visual Representation of a Biological Concept or Process

UNIT 7

Natural Selection

~19-21 Class Periods **13-20%** AP Exam Weighting

- 2** 7.1 Introduction to Natural Selection
- 1** 7.2 Natural Selection
- 4** 7.3 Artificial Selection
- 3** 7.4 Population Genetics
- 1** **5** 7.5 Hardy-Weinberg Equilibrium
- 4** 7.6 Evidence of Evolution
- 6** 7.7 Common Ancestry
- 3** 7.8 Continuing Evolution
- 2** 7.9 Phylogeny
- 2** **6** 7.10 Speciation
- 6** 7.11 Variations in Populations
- 3** 7.12 Origins of Life on Earth

Progress Check 7

Multiple-Choice: ~48 questions (2 parts)

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results with Graphing
- Analyze Data

UNIT 8

Ecology

~19-21 Class Periods **10-15%** AP Exam Weighting

- 3** 8.1 Responses to the Environment
- 6** 8.2 Energy Flow Through Ecosystems
- 4** 8.3 Population Ecology
- 5** 8.4 Effect of Density on Populations
- 5** 8.5 Community Ecology
- 6** 8.6 Biodiversity
- 5** 8.7 Disruptions in Ecosystems

Progress Check 8

Multiple-Choice: ~24 questions

Free-Response: 2 questions

- Interpreting and Evaluating Experimental Results with Graphing
- Scientific Investigation